

Menstrual disorders & Contraception

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Objective

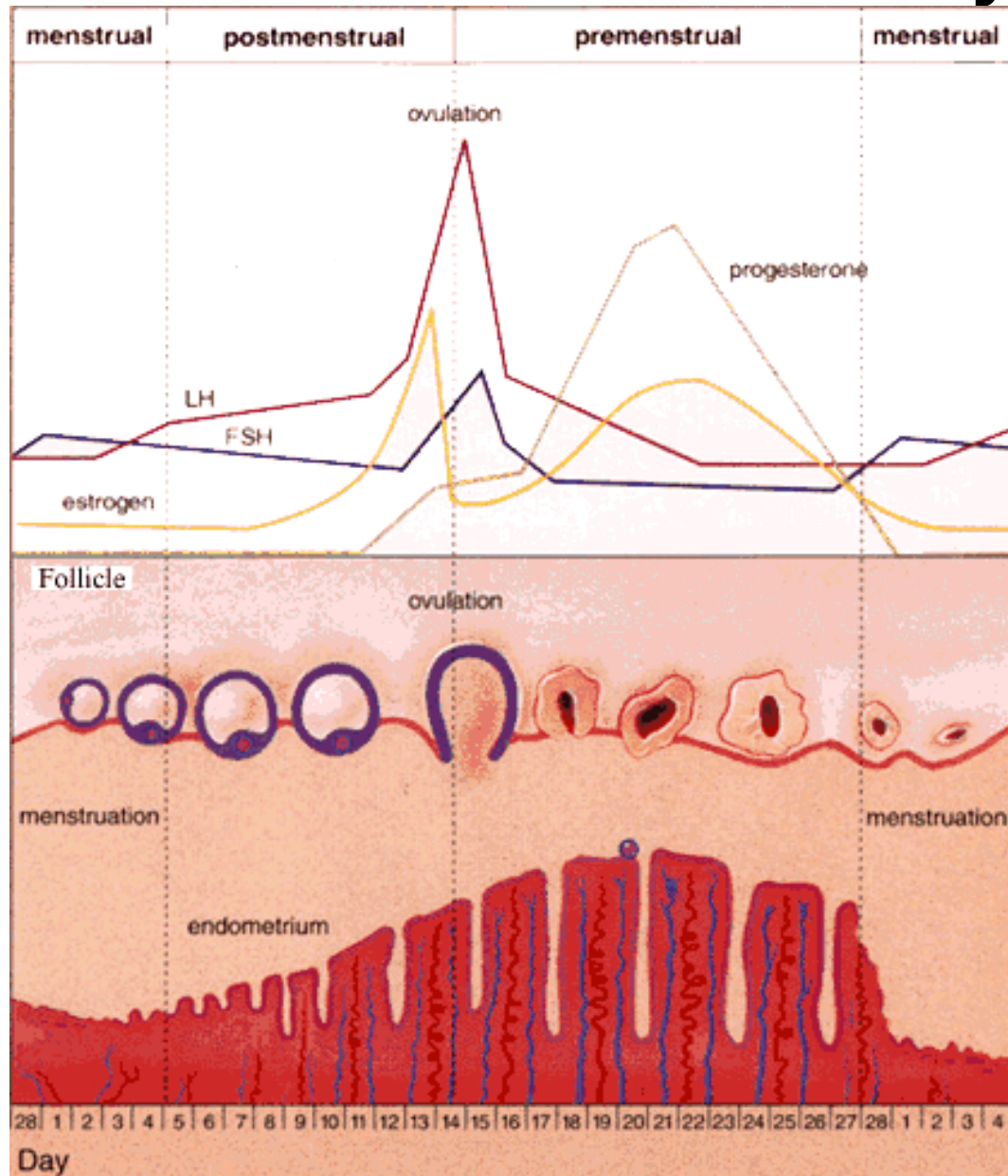
- Describe common menstrual disorders like abnormal uterine bleeding, amenorrhea, dysmenorrhea, and endometriosis and explain the diagnostic approach for evaluating menstrual abnormalities including hormonal testing and imaging.

Disclosures

- Dr. Rafferty is a certified Nexplanon trainer contracted employee of Organon

Menstrual Disorders

Normal Menstrual Cycle



Follicular development →
Ovulation (d14) →
Corpus luteal function →
Luteolysis

Endometrium is exposed to:
Ovarian production of estrogen →
(*proliferation*)

Combination of estrogen &
progesterone →
(*secretory phase*)

Estrogen & progesterone
withdrawal
(*desquamation and repair*)

Ovulation & Bleeding

Ovulatory Bleeding

- Normal ovulatory cycles
- Predictable pattern

Anovulatory Bleeding

- Ovulation failure which prevents corpus lutes formation & thus progesterone secretion
- Endometrium proliferates due to unopposed estrogen as there is no progesterone withdrawal bleeding
- Unpredictable freq & flow
- Example causes – PMOS, HypoThyroid, Perimenarche, Perimenopause, low BMI, Extreme Exercise

Abnormal Uterine Bleeding (AUB) Definitions - Less

- Amenorrhea – Absence of menstruation after menarche
 - Primary & Secondary
- Hypomenorrhea – Abnormally light menstrual flow
- Menorrhagia
- Menometrorrhagia
- Metrorrhagia
- Oligomenorrhea
- Polymenorrhea
- Post Menopausal Bleeding

Amenorrhea

Primary

- Menarche has not occurred by
 - age 15 with normal growth/body changes
 - age 13 with no signs of puberty

Secondary

- Cessation of previously
 - regular menses for 3+ mon
 - irregular menses for 6+ mon

Causes of Amenorrhea

TABLE 1

Select Causes of Amenorrhea*

Outflow tract abnormalities

Acquired

- Cervical stenosis
- Intrauterine adhesions

Congenital

- 5α-reductase deficiency
- Androgen insensitivity syndrome
- Imperforate hymen
- Müllerian agenesis
- Transverse vaginal septum

Primary ovarian insufficiency

Acquired

- Autoimmune
- Chemotherapy or radiation

Congenital

- Gonadal dysgenesis (other than Turner syndrome)
- Turner syndrome or variant

Hypothalamic or pituitary disorders

- Autoimmune disease
- Brain radiation

Hypothalamic or pituitary disorders (continued)

- Constitutional delay of puberty
- Empty sella syndrome
- Functional (overall energy deficit or stress)
- Eating disorder
- Stress
- Vigorous exercise
- Weight loss
- Gonadotropin deficiency (e.g., Kallmann syndrome)
- Hyperprolactinemia
- Adenoma (prolactinoma)
- Chronic kidney disease
- Medications or illicit drugs (e.g., antipsychotics, opiates)
- Physiologic (pregnancy, stress, exercise)
- Infarction (e.g., Sheehan syndrome)
- Infiltrative disease (e.g., sarcoidosis)
- Infection (e.g., meningitis, tuberculosis)
- Medications or illicit drugs (e.g., cocaine)
- Trauma or surgery
- Tumor (primary or metastatic)

Other endocrine gland disorders

- Adrenal insufficiency
- Androgen-secreting tumor (e.g., ovarian or adrenal)
- Cushing syndrome
- Diabetes mellitus, uncontrolled
- Late-onset congenital adrenal hyperplasia
- Polycystic ovary syndrome (multifactorial)
- Thyroid disease

Amenorrhea attributed to chronic disease

- Celiac disease
- Inflammatory bowel disease
- Other chronic disease

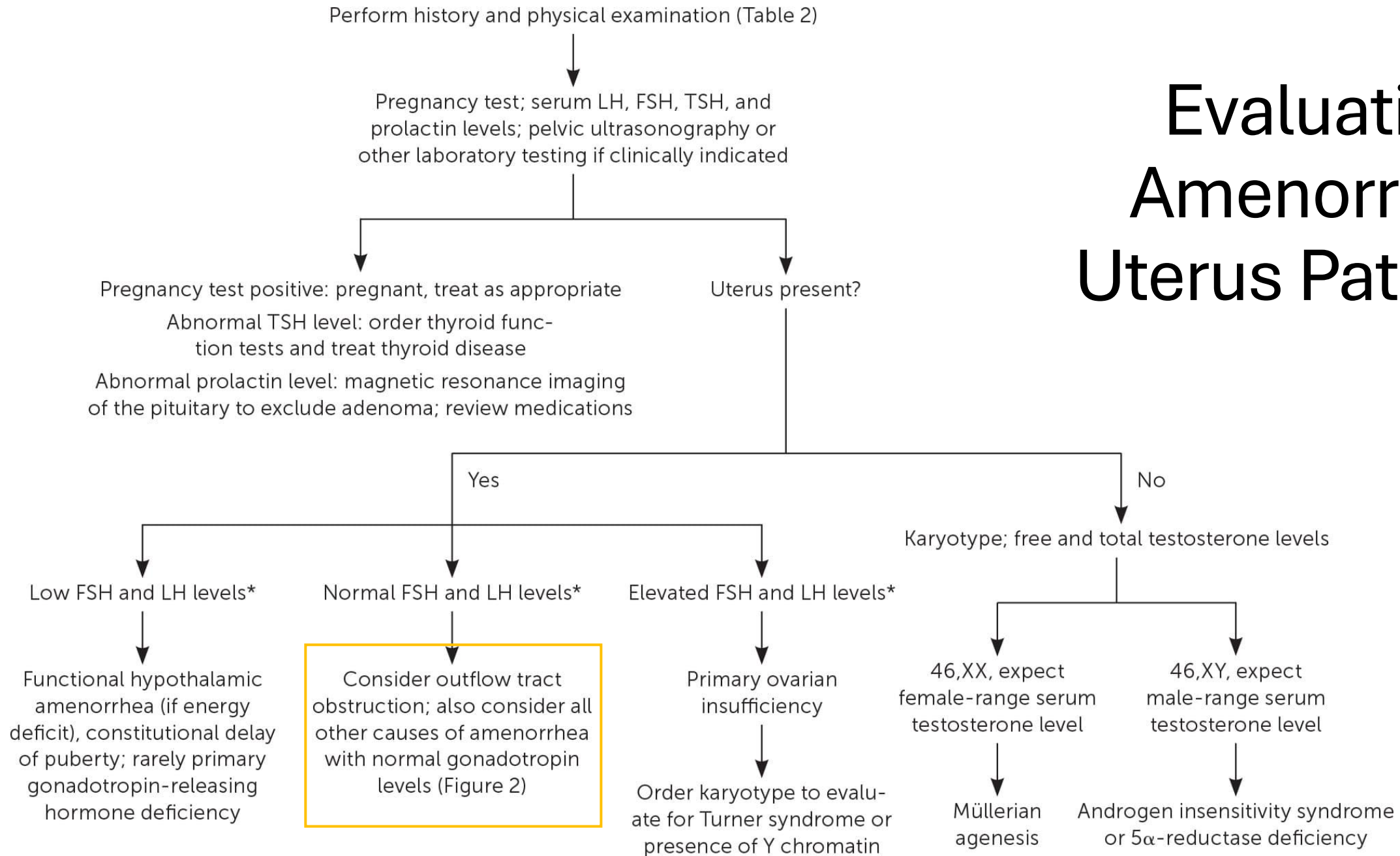
Physiologic or induced

- Breastfeeding
- Contraception
- Exogenous androgens
- Menopause
- Pregnancy

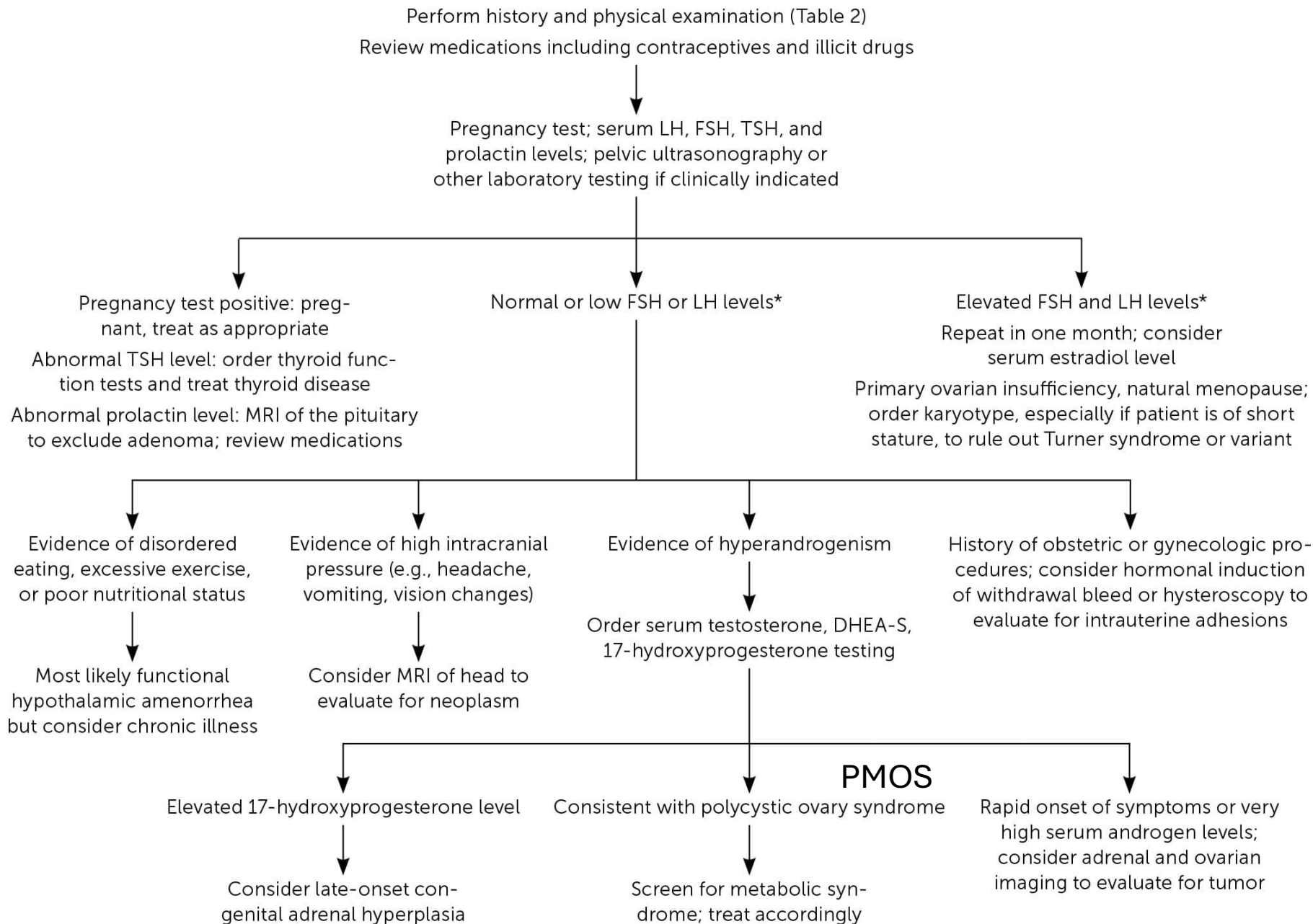
*—Conditions may span multiple categories.

Adapted with permission from Klein DA, Poth MA. Amenorrhea: an approach to diagnosis and management. Am Fam Physician. 2013;87(11):782, with additional information from references 1, 2, 5, 6, and 10 through 12.

Evaluation of Amenorrhea – Uterus Pathway



Evaluation of Amenorrhea – Hormonal Pathway



Amenorrhea

- Lab Work

TABLE 4

Interpretation of Common Laboratory Studies in the Evaluation of Amenorrhea*

	17-OHP	AMH	DHEA-S	Estradiol	LH (IU per L)	LH/FSH	FSH (IU per L)	Prolactin	Testosterone	TSH
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Polycystic ovary syndrome PMOS	Normal	Normal or high	High normal	Low	< 15	> 1	< 10	High normal	High or high normal	Normal
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17-OHP = 17-hydroxyprogesterone; AMH = anti-Müllerian hormone; DHEA-S = dehydroepiandrosterone-sulfate; FSH = follicle-stimulating hormone; LH = luteinizing hormone; TSH = thyroid-stimulating hormone.

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†—LH and FSH may be normal in functional hypothalamic amenorrhea.

Adapted with permission from Gordon CM, Ackerman KE, Berga SL, et al. Functional hypothalamic amenorrhea: an Endocrine Society clinical practice guideline. J Clin Endocrinol Metab. 2017;102(5):1423.

Primary Ovarian Insufficiency

- Secondary Amenorrhea occurring in patients <40 yrs old
- Follicle dysfunction or depletion
- Incidence – affects 1/100 females
- Symptoms - May have menopause symptoms (ie vasomotor symp, vaginal dryness)
- Causes – idiopathic, chemo, infections, tumors, autoimmune, chromosomal
- Diagnosis – lab work (ie FMR1 gene [↑ risk fragile X], thyroid, etc)
 - 2 serum FSH levels in menopause range (>25) at least 1 mon apart
- Treatment – HRT at least until natural menopause (50-51 yo)
- Complications – Bone Health, CV Risk, Dementia

TABLE 4

Amenorrhea - Lab Work POI

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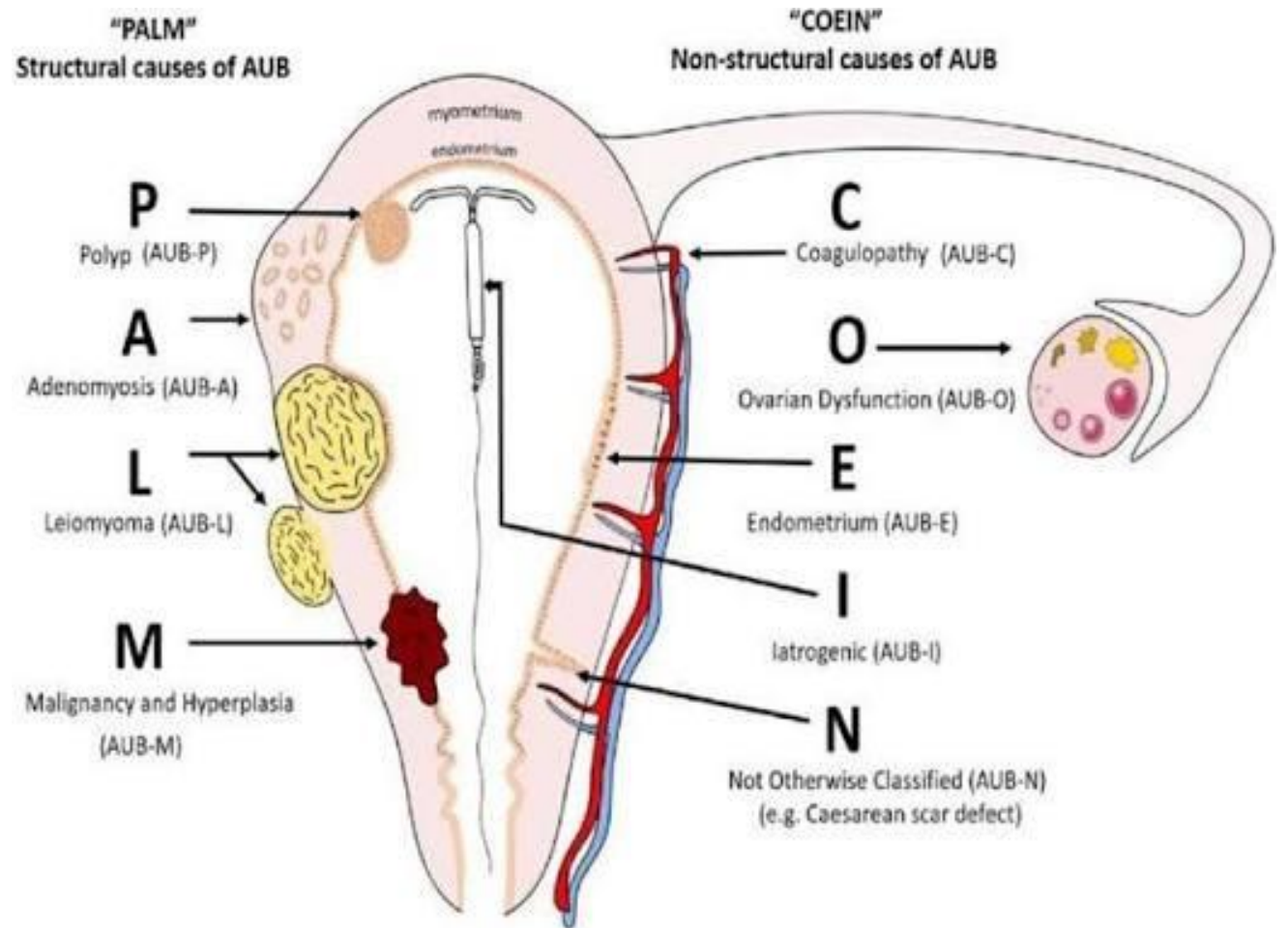
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Abnormal Uterine Bleeding Definitions - More

- Amenorrhea
- Hypomenorrhea
- Menorrhagia – Excessively prolonged menstrual flow, but regular
- Menometrorrhagia – Heavy menstrual flow that is irregular
- Metrorrhagia – Menstrual flow at irregular intervals unrelated to normal menses
- Oligomenorrhea
- Polymenorrhea
- Post Menopausal Bleeding

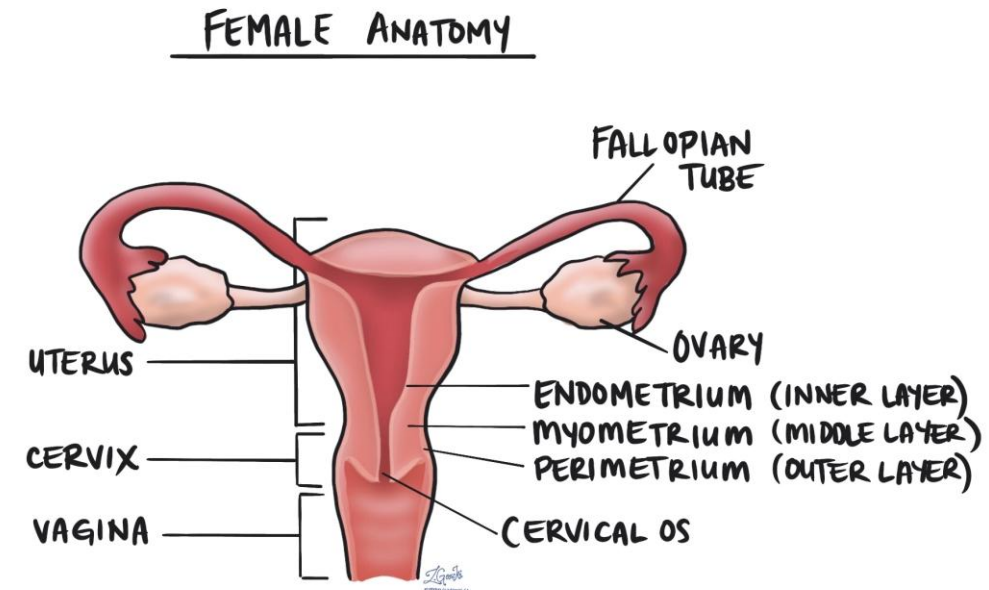
Abnormal Uterine Bleeding (AUB) Definitions

- Basically, any deviation from typical flow & frequency
- Causes can be structural or non-structural
 - PALM-COEIN



Structural Causes of AUB

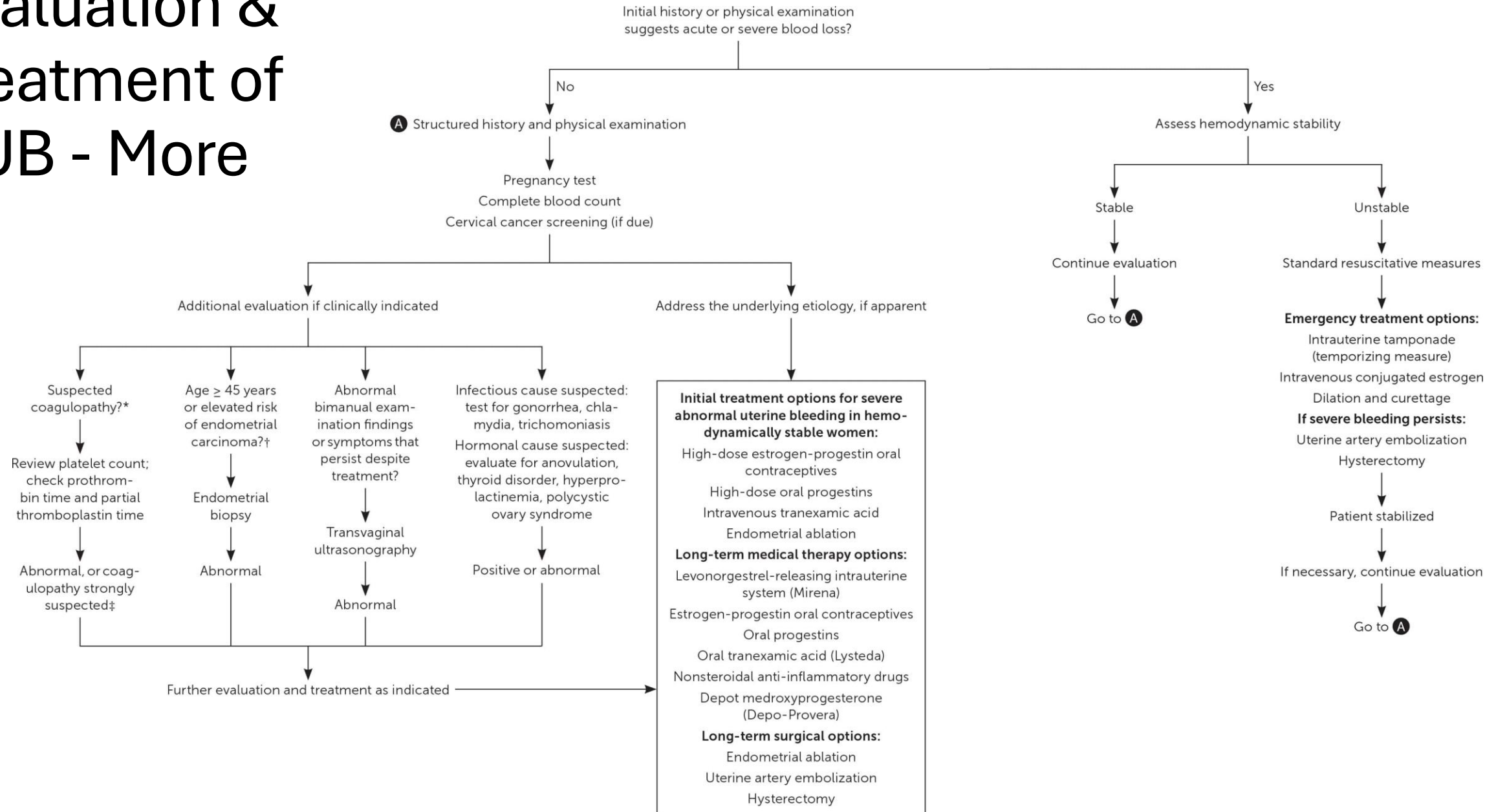
- Polyps – benign localized overgrowth of endometrial glands & stroma
- Adenomyosis – benign presence of endometrial glands & stroma w/in the myometrium of the uterus (not where they are supposed to be)
- Leiomyoma (aka fibroids) – benign smooth muscle tumors of the myometrium
- Malignancy & Endometrial Hyperplasia (precancer) – abnormal proliferation of endometrial tissue
 - Usually due to unopposed estrogen
 - Must be ruled out for post menopausal bleeding



Non-structural Causes of AUB

- Coagulopathy – a disorder of one of the components of coagulation
- Ovulatory dysfunction – a disorder of hormonal secretion from the ovary (ie Polyendocrine Metabolic Ovarian Syndrome, etc) or other hormonal disorder that affects ovarian function (ie thyroid disorders, diabetes, prolactinomas, etc)
- Endometrial – a dysfunction of the endometrial lining despite normal hormonal secretion
- Iatrogenic – causes related to medications or medical procedures (ie hormonal contraceptive therapies, anticoagulants, etc)
- Not Otherwise Classified – causes are poorly understood or rare (AV malformations, surgical scars, etc)

Evaluation & Treatment of AUB - More



Abnormal Uterine Bleeding Definitions - Freq

- Amenorrhea
- Hypomenorrhea
- Menorrhagia
- Menometrorrhagia
- Metrorrhagia
- Oligomenorrhea – Infrequent menstrual flow (>35 days)
- Polymenorrhea – Menstrual cycles occurring too frequently (<21 days)
- Post Menopausal Bleeding

Polyendocrine Metabolic Ovarian Syndrome (PMOS)

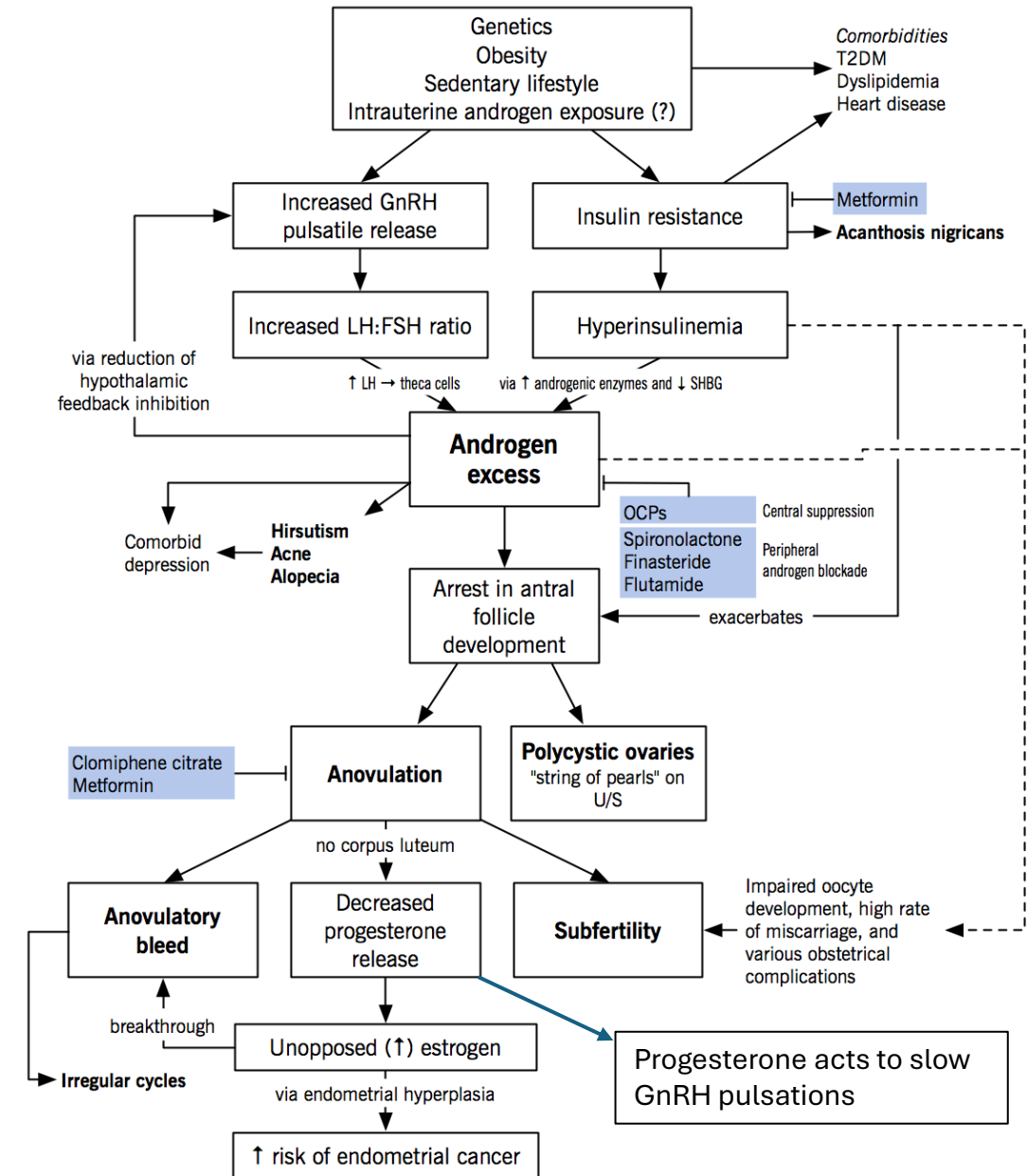
- Primary or Secondary Amenorrhea; Oligomenorrhea
- Incidence: most common endocrinopathy in repro aged women affecting 13% of this population
- Symptoms – Facial Hair, Acne, Male Pattern Baldness, Central Obesity, Menstrual Abnormalities, Acanthosis Nigricans
- Causes – multifactorial (genetic, environmental, testosterone exposure in utero)



PMOS (PCOS) Pathophysiology

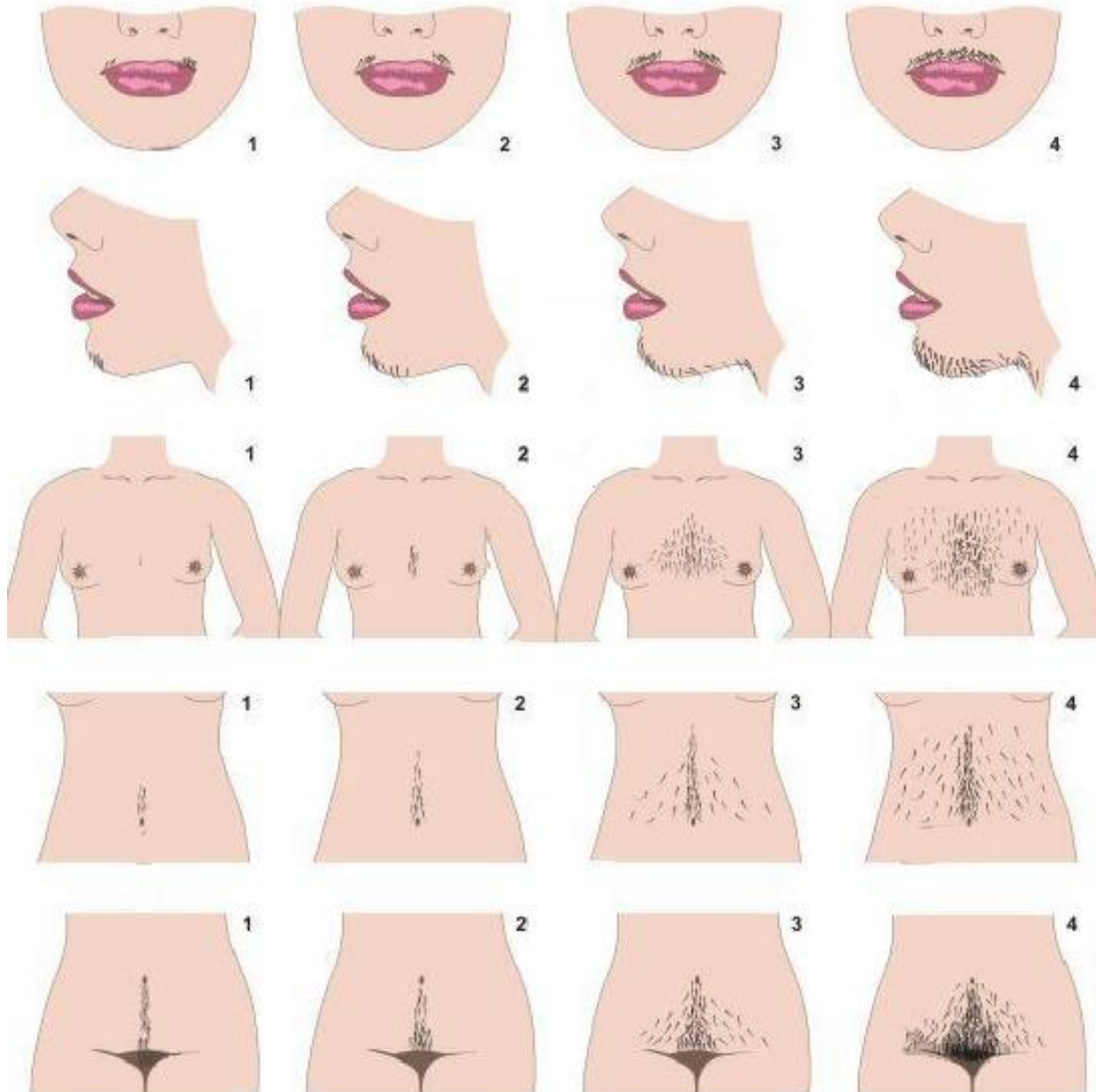
Pathophysiology of PCOS

Alex Rotstein, Ragini Srinivasan, and Eric Wong



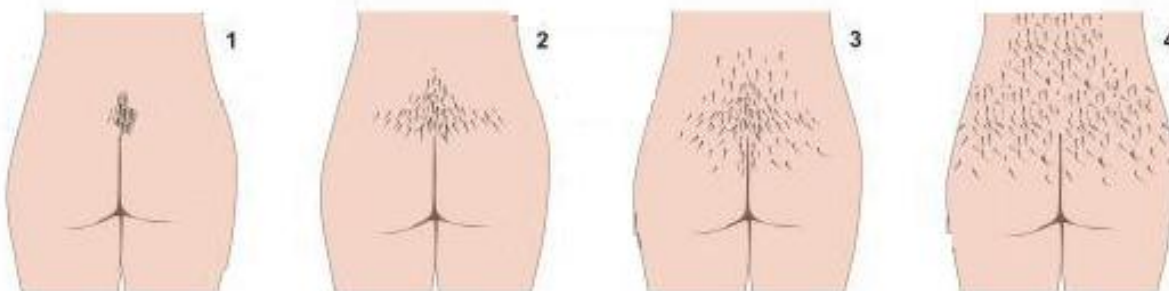
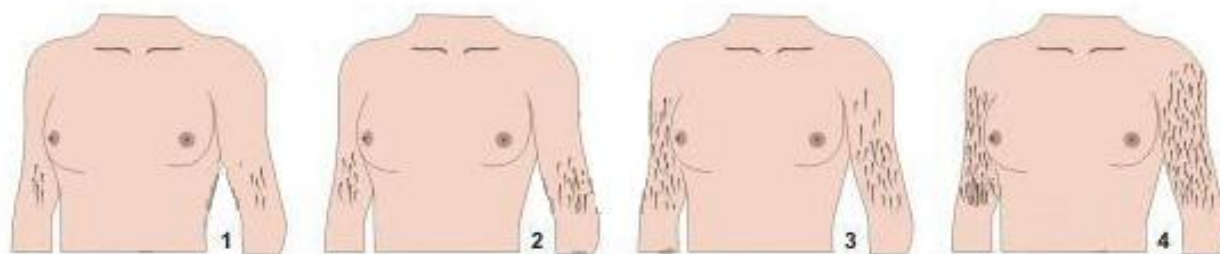
PMOS (formerly PCOS)

- Diagnosis – lab work (ie thyroid, LH/FSH, AMH, Testosterone, etc)
 - Rotterdam Criteria (meet 2/3)
 - Clinical or Biochemical Hyperandrogenism
 - Facial & Terminal Hair Growth as evaluated by the Ferriman-Gallwey System
 - Testosterone (total & free)
 - Ovulatory Dysfunction
 - Amenorrhea or Oligomenorrhea
 - Progesterone levels
 - Polycystic ovarian morphology – Pelvic ultrasound
 - Pelvic ultrasound not required, but if obtained
 - Excessive size of ovary $>10.5 \text{ mL}^3$ in adults
 - Excessive number of follicles $> 10/\text{plane}$ or $>12/\text{ovary}$ in absence of dominant-size follicle of $>1.0 \text{ cc}$ or corpus luteum
- Treatment – Hormonal Therapy (OCP), Diet/Exercise, Weight Loss, Metformin, Spironolactone, Letrozole, Clomiphene Citrate, Ovarian Diathermy, IVF
- Complications – Metabolic (Insulin Resistance/DM, Obesity, Hyperlipidemia, Hypertension, NASH), Infertility, Cancer (Endometrial, Ovarian)



Ferriman-Gallwey System

<http://www.hirsutism.com/hirsutism-biology/ferriman-gallwey-score.shtml>



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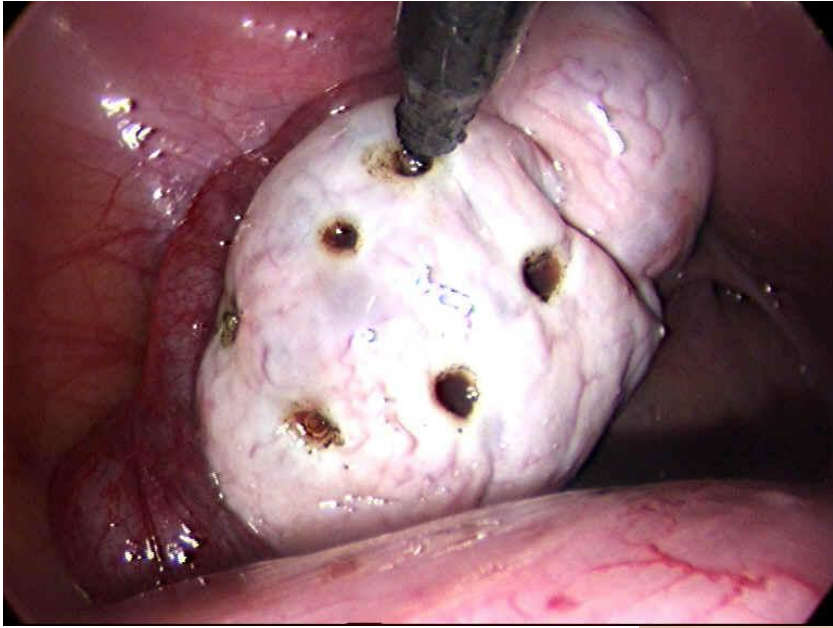
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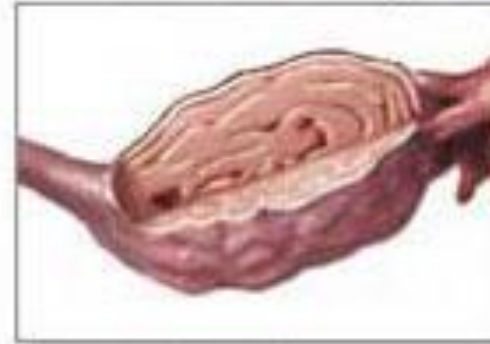
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PMOS - Images



Normal ovary



Polycystic ovary



Ultrasound of Ovaries



Normal Ovary



PCOS Ovary

ADAM.

http://tiredofwaiting.files.wordpress.com/2007/07/picture_ovaries.jpg

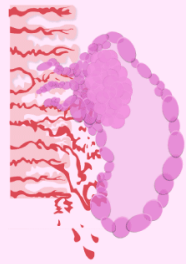
http://www.bio.davidson.edu/people/vecase/seniorcolloquium/04/Ovarian_Drilling.jpg

Abnormal Uterine Bleeding Definitions – Post Menopause

- Amenorrhea
- Hypomenorrhea
- Menorrhagia
- Menometrorrhagia
- Metrorrhagia
- Oligomenorrhea
- Polymenorrhea
- Post Menopausal Bleeding – Any uterine bleeding that occurs after menopause (defined as the time 1 yr after last menses)

Causes of AUB

- Always remember pregnancy!



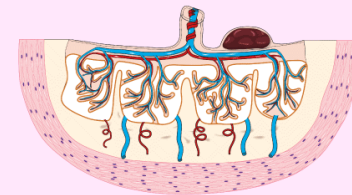
Implantation
bleeding



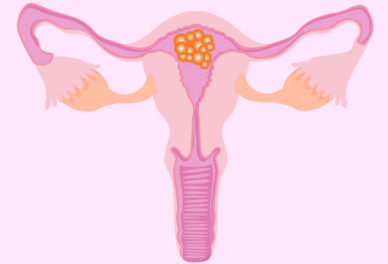
Spontaneous
abortion



Ectopic
pregnancy



Intrauterine
hematoma



Molar
pregnancy

Is bleeding in pregnancy normal? What are the causes?

Differential Diagnosis for AUB

- Vaginal infections – STIs, yeast
- Vulvar conditions – lichen, ulcer/sore, hymenal
- Post-coital – cervical/vaginal trauma, post-menopausal, hymenal
- Cervical – post pap/colposcopy procedure, STIs
- Urinary - UTIs, urethra trauma (foley?), kidney stone
- Uterine – polyps, fibroids, endometrial, malignancy
- Ovarian & HPG axis – PMOS, prolactinoma, primary ovarian insufficiency, perimenopause/menopause
- Pregnancy – ectopic, miscarriage, molar, implantation
- GI conditions – rectal trauma, hemorrhoids, GI bleed
- Extra HPG axis hormonal abnormalities – thyroid condition
- Hematologic – Coagulation disorder
- Medications – hormonal contraception, anticoagulants

eTABLE 2

Differential Diagnosis of Abnormal Uterine Bleeding

Uterine bleeding

Coagulopathies

Factor deficiencies
Platelet dysfunction
Thrombocytopenia
von Willebrand disease

Iatrogenic

Anticoagulants
Antidepressants
Antipsychotics
Chemotherapeutic agents
Copper intrauterine device
Corticosteroids
Hormonal contraception or other hormone therapy
Phenytoin (Dilantin)
Tamoxifen

Infection

Acute or chronic endometritis
Pelvic inflammatory disease

Ovulatory dysfunction

Androgen excess
Androgen insensitivity syndrome
Hormone-producing tumors
Polycystic ovary syndrome
Hypothalamic-pituitary-adrenal axis dysfunction
Congenital adrenal hyperplasia
Cushing syndrome/disease
Hyperprolactinemia
Immature hypothalamic-pituitary-adrenal axis (adolescence)
Intense exercise, stress
Lactational amenorrhea
Ovarian follicle decline (perimenopause)
Starvation (including eating disorders)
Thyroid disorders
Tumors, radiation, or trauma of the pituitary/hypothalamus area
Premature ovarian failure

Pregnancy

Abortion
Abruptio
Ectopic pregnancy
Gestational trophoblastic disease
Subchorionic hemorrhage

Structural

Adenomyosis
Arteriovenous malformations
Cesarean scar defect
Endometriosis
Leiomyomas
Malignancy
Endometrial hyperplasia/carcinoma
Metastasis
Uterine sarcoma
Outflow obstruction
Polyp

Nonuterine bleeding

Adnexa

Malignancy
Pelvic inflammatory disease

Anus

Anal fissure
Hemorrhoids
Inflammatory bowel disease
Upper or lower gastro-intestinal bleeding

Bladder/ureters/kidneys

Infection
Malignancy
Nephrolithiasis

Cervix

Dysplasia/malignancy
Ectropion
Endometritis
Infection (e.g., gonorrhea, chlamydia)
Polyps
Urethra
Urethral diverticula
Urethral prolapse
Urethritis

Vagina

Atrophy
Benign growths
Infection
Malignancy
Retained foreign body
Trauma
Ulcerative conditions
Vulva
Benign growths
Blistering diseases
Malignancy
Trauma

Other

Chronic kidney disease
Chronic liver disease
Diabetes mellitus
Leukemia
Sarcoidosis of the reproductive tract
Tuberculosis of the reproductive tract
Turner syndrome

Evaluation of AUB

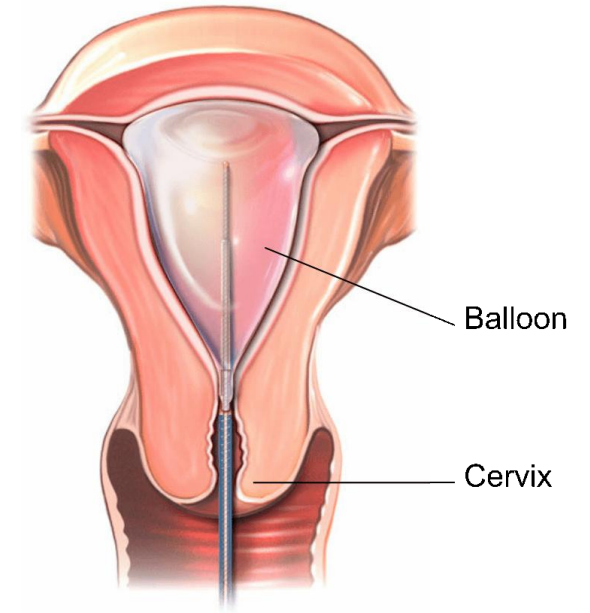
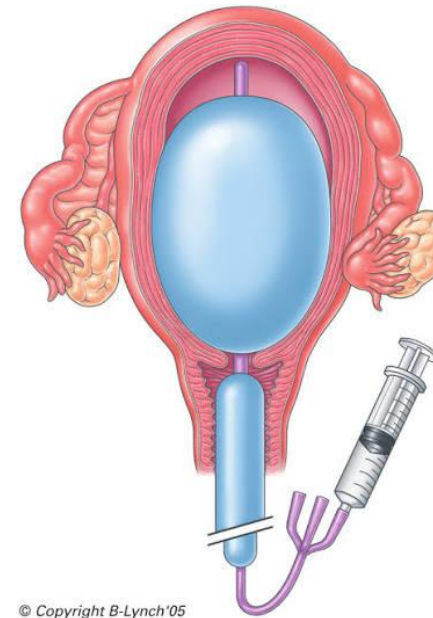
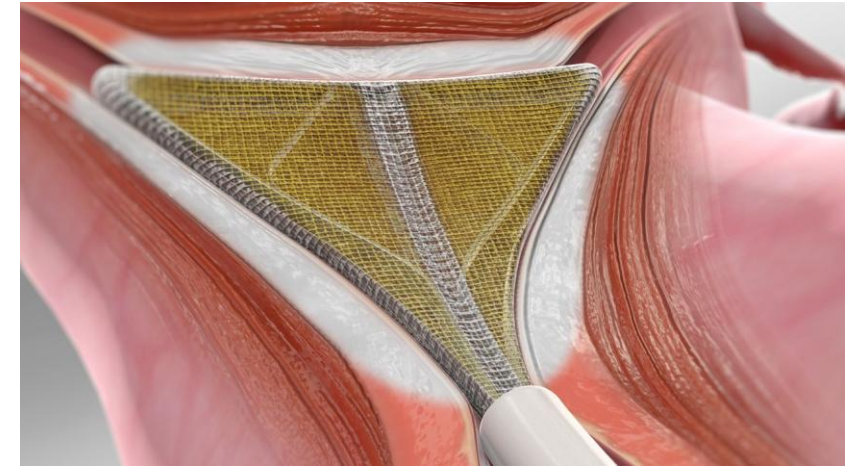
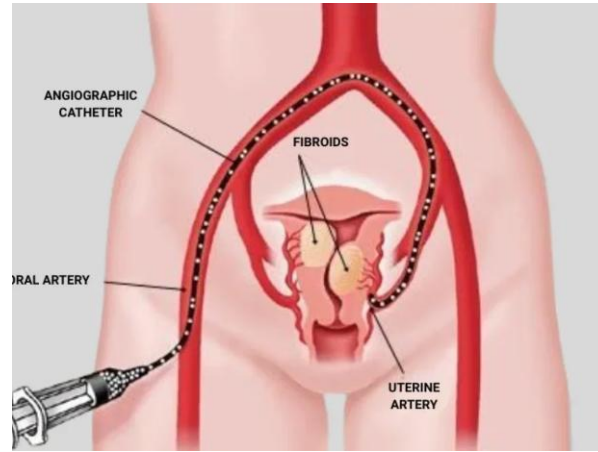
- History – 80%
 - Menstrual History
 - Bleeding Pattern
 - Assessment of Pain
 - Sexual History
 - Associated Symptoms
- Physical Exam – 10-15%
 - General – BMI, obesity
 - Skin – acne, bruising, acanthosis nigricans, hair distribution
 - Neck – Thyroid
 - Abdominal
 - Pelvic
- Labs & Imaging – 5-10%

Labs & Imaging Evaluation for AUB

- Infection swabs – yeast, STIs
- Urine pregnancy test or Serum B-hcg
- CBC, PTT/INR, Coag Panel, TSH, Prolactin, CMP, A1C, LH/FSH, DHEA-S, Testosterone, AMH
- Pap smear
- Pelvic Ultrasound – including transvaginal
 - Endometrial stripe is very important esp in postmenopausal bleeding evaluation & probably of endometrial cancer (<4mm)
- Endometrial Sampling – endometrial biopsy vs Dilation & Curettage
- Hysteroscopy

Treatment for AUB

- Expectant Management!
- Medications
 - Hormonal Contraceptives
 - Tranexamic Acid
- Procedures/Surgeries
 - Uterine tamponade
 - Uterine artery embolization
 - Endometrial ablation
 - Ovarian Diathermy
 - Myomectomy or polypectomy
 - Dilation & Curettage
 - Hysterectomy



[PPH 2nd edn #23.vp](#)

[Uterine Artery Embolisation in NZ: Safe and Minimally Invasive](#)

[Endometrial Ablation \(Radiofrequency Method\)](#)

[Thermal balloon endometrial ablation | healthdirect](#)

Dysmenorrhea

- Painful menstruation
- Incidence – 50-90%, only about 45% will present
 - Usually starts 6-12 mon following menarche
- Risk Factors – age <30, BMI <20, smoking, younger menarche, longer cycles, heavier flow, and h/o sexual abuse
- Protective Factors – age >30, increasing parity, exercise, hormonal usage like OCPs
- MOA – Elevated Prostaglandin & Leukotriene levels with inflammation causing uterine contractions & cramping

Dysmenorrhea - Secondary Causes

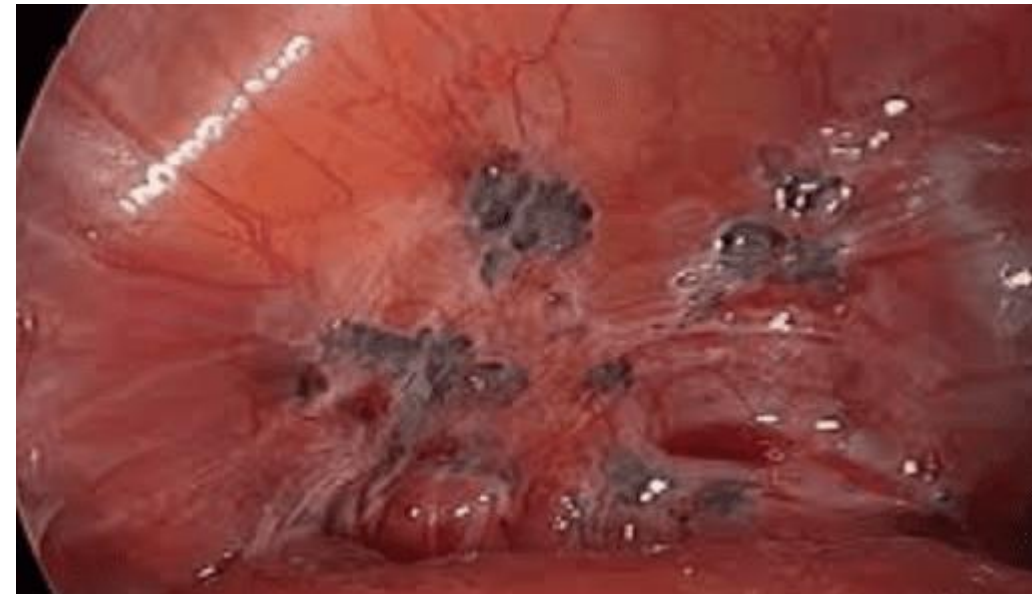
- Endometriosis
- Adenomyosis
- Fibroids
- Congenital Anomaly (ie mullerian malformations)
- Pelvic Adhesions (ie prior surgery)
- Pelvic Inflammatory Disease

Dysmenorrhea Eval & Treatment

- Symptoms – cramping pain in the lower abdomen/pelvis beginning at onset of menstrual flow & lasting 8-72 hours
 - Primary associated symp: nausea, vomiting, diarrhea, headaches, muscle cramps, low back pain, fatigue, sleep disturbance
 - Secondary associated symp: abnormal uterine bleeding, vaginal discharge, dyspareunia
- Diagnosis – history, abdominal/pelvic exam
 - Labwork (ie urine, STI/Vaginosis testing, pregnancy, etc)
 - Pelvic ultrasound
 - Laparoscopy
- Treatment – NSAIDs, Hormonal Therapy (OCP, leuprolide acetate), Exercise (45-60 min 3x/wk), TENs, Heat, Acupuncture, Surgery
- Complications – Mood disorders, Missed Work/School, Infertility

Endometriosis

- Presence of endometrial glands & stroma in extrauterine locations
 - Estrogen promotes implantation of endometrial tissue & stimulates inflammation
 - Neurogenesis, which is indicated by high levels of nerve growth factors that alter & increase sensory and sympathetic nerve fibers, is observed in endometriotic tissue.
- Prevalence – peak is 25-35 yo
- Incidence – most common gyne disorder, affecting 10% repro age pts
- Risk Factors – increased flow, low BMI, FH, nulliparity, short cycles, early menarche



Endometriosis Eval & Treatment

- Symptoms
 - Dysmenorrhea (painful) & Menometrorrhagia (heavy/irregular)
 - Dyspareunia
 - Chronic pelvic pain
 - Infertility
 - Cyclical GI symp (ie painful bowel movements)
 - Cyclical urinary symp
- Diagnosis – delayed 4-10 yrs
 - History & Physical exam – abdominal/pelvic
 - Imaging limitations
 - Pelvic Ultrasound (85% specificity, 50-84% sensitivity)
 - MRI of pelvis +/- abdomen w/o contrast
 - Laparoscopy
- Management
 - Medications
 - Hormonal – suppress estrogen (OCPs, Depo/Nexplanon, etc but less so IUDs), leuprolide acetate
 - Analgesics – NSAIDs
 - Aromatase inhibitors – Letrozole, etc
 - Alternative treatments – acupuncture, exercise, dietary changes (avoid gluten & dairy)
 - Laparoscopy – ablation of lesions

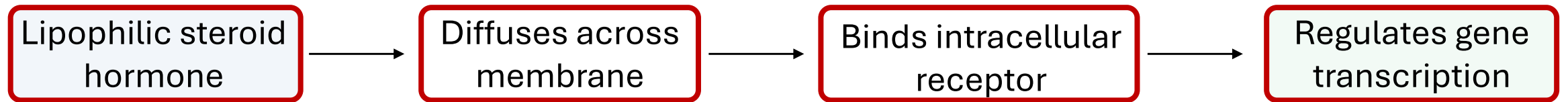
Reproductive Pharmacology

Estrogen & Progestogens

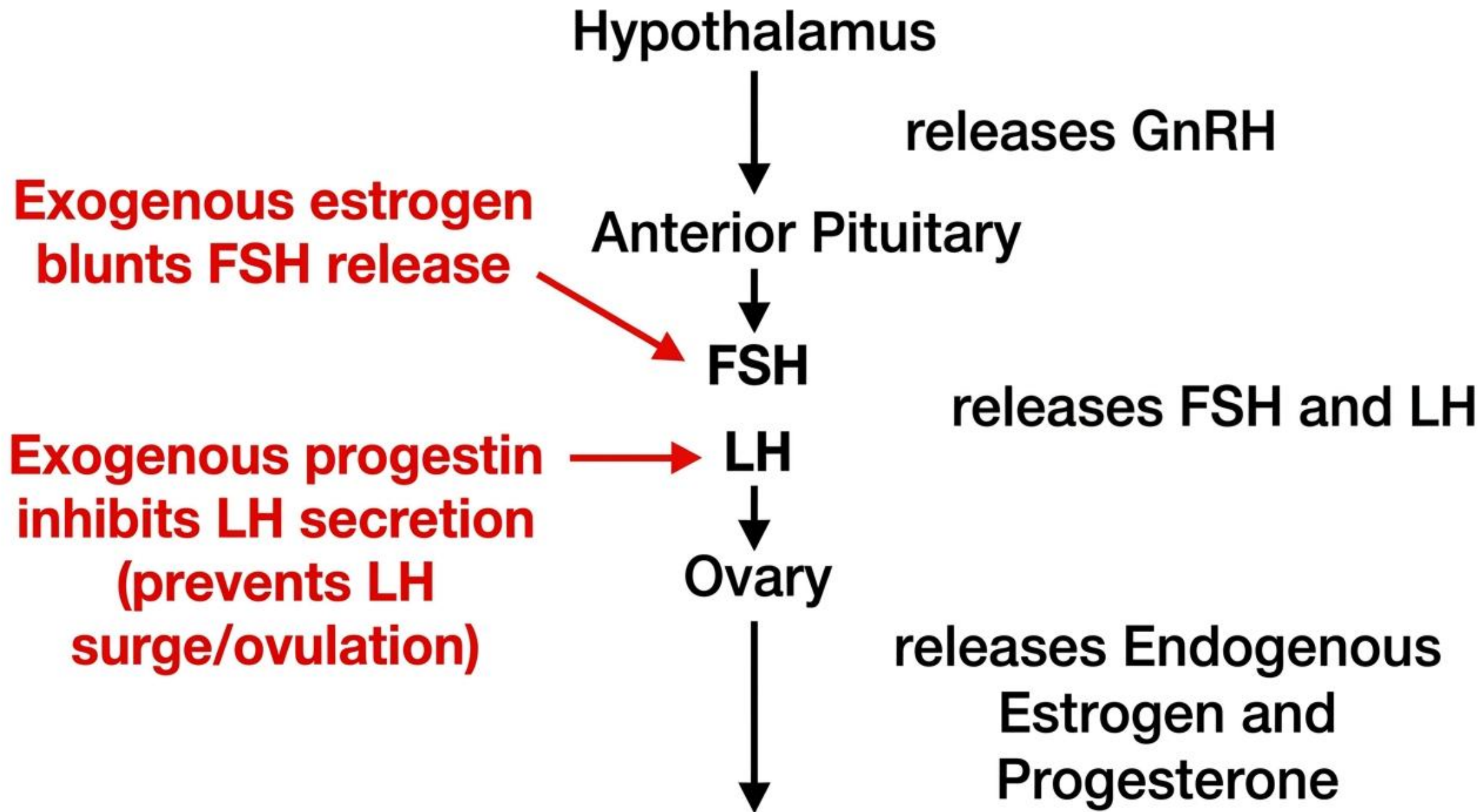
Session Objectives

- Compare hormonal contraceptive therapies in terms of mechanism of action, efficacy, indications, contraindications, and adverse effects
- Explain the pharmacology of medications used for emergency contraception and induced abortion, including their mechanism of action and adverse effects

Steroid Hormone Pharmacology



N



Contraceptive Mechanism: Estrogen vs Progestin

Estrogen component

- Negative feedback at hypothalamus/pituitary
- Blunts FSH release
- Prevents development of a dominant follicle
- Stabilizes endometrium and improves cycle control

Progestin component

- Inhibits LH surge → prevents ovulation
- Thickens cervical mucus → slows sperm transport
- Suppresses/atrophies endometrium
- alters/suppresses the endometrium

Hormonal Contraceptive Classes

Combined hormonal contraception

Oral pill
Transdermal patch
Vaginal ring

Progestin-only contraception

Progestin-only pill
DMPA injection
Etonogestrel implant
Levonorgestrel IUD

Emergency contraception

Levonorgestrel
Ulipristal

Method choice is often driven less by the molecule and more by route, adherence, estrogen risk, and bleeding profile.

Combined Oral Contraceptives (COCs)

Formulation and schedule

- Contain estrogen, most often ethinyl estradiol, plus a progestin.
- Monophasic: constant estrogen/progestin dose.
- Triphasic or multiphasic: changing hormone doses across cycle-varying progestin doses to mimic cycle
- Typical cycle: active hormone pills followed by hormone-free interval.

Notes

- Suppress gonadotropins → prevent ovulation.
- Withdrawal bleeding occurs during placebo/hormone-free interval.
- Extended-cycle or continuous use reduces frequency of withdrawal bleeding.
- Useful for cycle regulation as well as contraception.

Combined Oral Contraceptives (COCs)--

Method	Contains	Administration	Notes
Combined Oral Contraceptives (COCs)	Ethinyl estradiol + progestin	21-24 days active pills, 4-7 days placebo	Monophasic: constant estrogen/progestin dose Vs Triphasic or multiphasic: changing hormone doses across cycle varying progestin does to mimic cycle

Patch and Vaginal Ring

Method	Contains	Administration	Notes
Transdermal patch	Ethinyl estradiol + progestin (Norelgestromin /Levonorgestrel)	Weekly patch for 3 weeks; 1 week off	Comparable efficacy to oral contraceptives; higher estrogen exposure- increase risk of VTE. less effective >90 kg , BMI >30
Vaginal ring	Ethinyl estradiol + etonogestrel	Ring in place for 3 weeks; 1 week off	Device-related discontinuation: irritation or expulsion

Pill, patch, and ring are estrogen-containing methods.
Contraindications to estrogen generally apply across the groups.

When to Avoid Estrogen-Containing Contraception

Contraindications

- Smoking >35,
- Cerebrovascular/thromboembolic disease,
- Estrogen-dependent neoplasms,
- Liver disease
- Pregnancy

Notes

If estrogen risk is unacceptable
→ pivot to progestin-only or nonhormonal options.

Current breast cancer: avoid hormonal contraception broadly; use nonhormonal options.

Progestin-Only Contraception

Shared progestin effects

Suppress LH and ovulation to variable degrees.

Thicken cervical mucus.

Suppress/alter endometrium.

Do not carry the same estrogen-related thrombotic risk pattern.

Shared adverse-effect pattern

Irregular bleeding is common, especially early.

Headache, mood symptoms, weight gain, and libido changes can occur.

Androgenic progestins can worsen acne/hirsutism.

Choice often depends on adherence needs and bleeding preferences.

Adverse Effects: Split by Hormonal Driver

Hormonal component	Common effects	Serious / distinctive risks
Estrogen	Nausea, breast tenderness, headache, fluid retention	Thromboembolism, hypertension, MI/stroke risk pattern
Progestins	Headache, mood symptoms/depression, weight gain, libido changes, irregular bleeding	Androgenic effects with some progestins; DMPA bone loss; drospirenone hyperkalemia risk

Most **high risk** “avoid this patient” warnings are driven by **estrogen**; most **method-specific** nuisance effects are driven by **progestin** delivery.

Progestin-Only Pills (POPs)

Useful when estrogen is undesirable

Pharmacology

Daily: Usually norethindrone in classic “mini-pill” formulations.

Low continuous dose.

Mechanism relies heavily on cervical mucus and variable ovulation suppression.

Drospirenone POP is an alternative formulation.

Clinical use and cautions

May be used in breastfeeding patients or those with estrogen contraindications.

Daily adherence matters.

Irregular bleeding is more frequent.

Rapid return to fertility after stopping classic POPs is a useful vignette clue.

Depot Medroxyprogesterone Acetate (DMPA)

High sustained progestin exposure

Mechanism and delivery

- IM/SC Medroxyprogesterone acetate injection **every 3 months**.
- Sustained progestin levels strongly suppress ovulation.
- Many users develop amenorrhea over time.

Distinctive adverse effects

- Weight gain is common.
- Return to fertility can be delayed for several months after stopping.
- Bone mineral density loss is the key long-term adverse effect.

Etonogestrel Implant

Long-acting reversible contraception (**LARC**)

Mechanism

- Subdermal etonogestrel release.
- Strong ovulation suppression.
- Thickened cervical mucus contributes.

Why it is high efficacy

- Lasts up to 3 years
- Efficacy does not rely on daily or weekly adherence.
- Reversible when removed.

Adverse-effect clue

Irregular
menstrual
bleeding
+ headache

Levonorgestrel-Releasing IUD

Local progestin effect with high efficacy

Mechanism

IUD-Levonorgestrel released inside uterus.

Thickens cervical mucus.

Alters endometrium and sperm function/survival.

Ovulation may continue in some users.

Clinical profile

Long-acting reversible contraception (3-8 yrs)
Appropriate when patient desires long-term contraception.

Highly useful for heavy menstrual bleeding.
Initial spotting can occur; bleeding often becomes lighter/amenorrheic.

Noncontraceptive Therapeutic Uses of Hormonal Contraceptives

- Heavy menstrual bleeding
- Dysmenorrhea
- Endometriosis
- May improve Acne-combined

Additional Benefit of **Combined** Oral Contraceptives

- Associated with ↓ risk of endometrial cancer
- Associated with ↓ risk of ovarian cancer

Efficacy-Adherence

Highest typical-use efficacy

LARC: implant + LNG-IUD

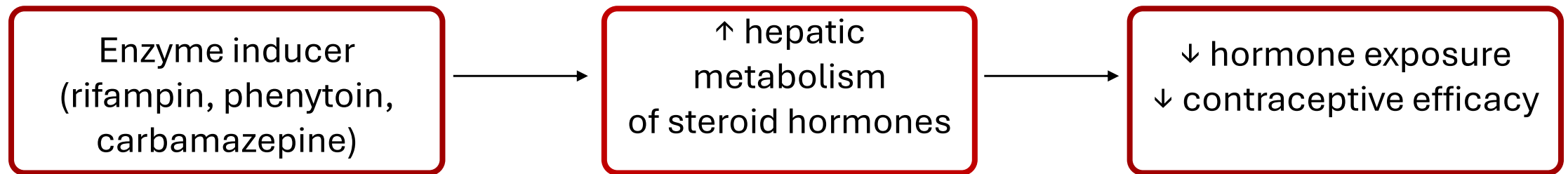
DMPA injection
Clinic/self injection every 3 months

Pill / patch / ring / POP
Daily, weekly, or monthly user action

Barrier/contextual methods
Use at intercourse

LARC therapeutic exposure is **maintained without**
relying on users' **adherence**

Drug Interactions: Enzyme Inducers



Emergency Contraception

Method	Class	Timing	Mechanism
Levonorgestrel	High-dose progestin	As soon as possible; preferably within 72 h	Primarily delays/prevents ovulation
Ulipristal acetate	Progesterone agonist/antagonist	Within 5 days	Delays ovulation; progesterone receptor modulation
Copper IUD	Nonhormonal	Within 5 days	Toxic to sperm/ova; prevents fertilization

Emergency contraception works primarily by preventing or delaying ovulation **and does not terminate an existing pregnancy**; mifepristone/misoprostol are used to terminate an intrauterine pregnancy.

Medications used for induced abortion

Mifepristone-Progesterone antagonist

Mechanism

Competitive progesterone-receptor antagonist.

Interferes with progesterone needed to maintain pregnancy.

Causes decidual breakdown/separation and cervical softening.

Increases uterine responsiveness to prostaglandins.

Adverse effects

Abdominal pain and uterine bleeding are expected pharmacologic consequences.

Treatment Failure

Incomplete termination can occur.

Mechanistic contraindication:

ectopic pregnancy is not treated by this regimen.

Also antagonizes glucocorticoid receptors → explains **contraindication** in chronic adrenal failure / long-term corticosteroid therapy.

Misoprostol-Prostaglandin analog

Mechanism

Prostaglandin E1 analog.
Stimulates myometrial contraction.
Promotes cervical ripening.
Used after mifepristone to actively expel pregnancy tissue.

Adverse effects

Cramping / abdominal pain.
Uterine bleeding.
Nausea, vomiting, diarrhea.
Fever/chills can occur due to prostaglandin effects.

Final Objective

- Develop a foundational understanding of discussing sensitive topics surrounding sexual health in adults and adolescents, how to discuss consent, and counseling on sexual health topics like contraception.

Sexual History & Counseling on Safe Sexual Practices

- 5 Ps System
 - Partners – Gender of preferred partner(s) & # of partners
 - Practices – Age at first coital encounter, risky sexual behaviors, consent
 - Prevention of STIs – Use of STI prevention (condoms, testing, vaccines)
 - PMH of STIs
 - Pregnancy Intentions – Use of Contraception

Contraceptive Counseling

- Understand patient's level of knowledge
- Consider patient practices & goals (avoid assumptions)
- Administration & Timing of different medication options
 - Preventative Contraception vs Emergency Contraception
- Efficacy of different medication options
- Patient risks/contraindications & drug interactions

Reproductive Health in the Care of Minors

- All states have laws that allow minors to consent to services surrounding sexual health, but states vary on age limitations
 - [Minors' Access to Contraceptive Services | Guttmacher Institute](#)
 - ‡Illinois does not have laws specific to minors' consent to contraception. However, they have laws that establish the right to contraception, establish noninterference by the state in contraceptive access, or explicitly permit a state entity to provide contraceptive care to any person who may benefit from it. These laws do not mention age specifically, nor has there been any case law establishing a minors' right to contraceptive services in these states.

Minors' Access to Contraceptive Services

Jurisdiction	Explicitly allows minors to consent to contraception	Allows minors to consent to contraception under specified circumstances:					Prohibits minors from accessing contraception without parental consent
		Health risk	Married or ever married	Parent	Pregnant or ever pregnant	Other	
Illinois ‡		X	X	X	X	X (with medical or other referral)	

Questions?

References - Rafferty

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- [Dysmenorrhea | AFP](#)
- [Endometriosis: Evaluation and Treatment | AFP](#)
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- Lippincott® Illustrated Reviews: Pharmacology, 8e Karen L. Whalen, Sarah M. Lerchenfeldt, Chris R. Giordano. Chapter 25